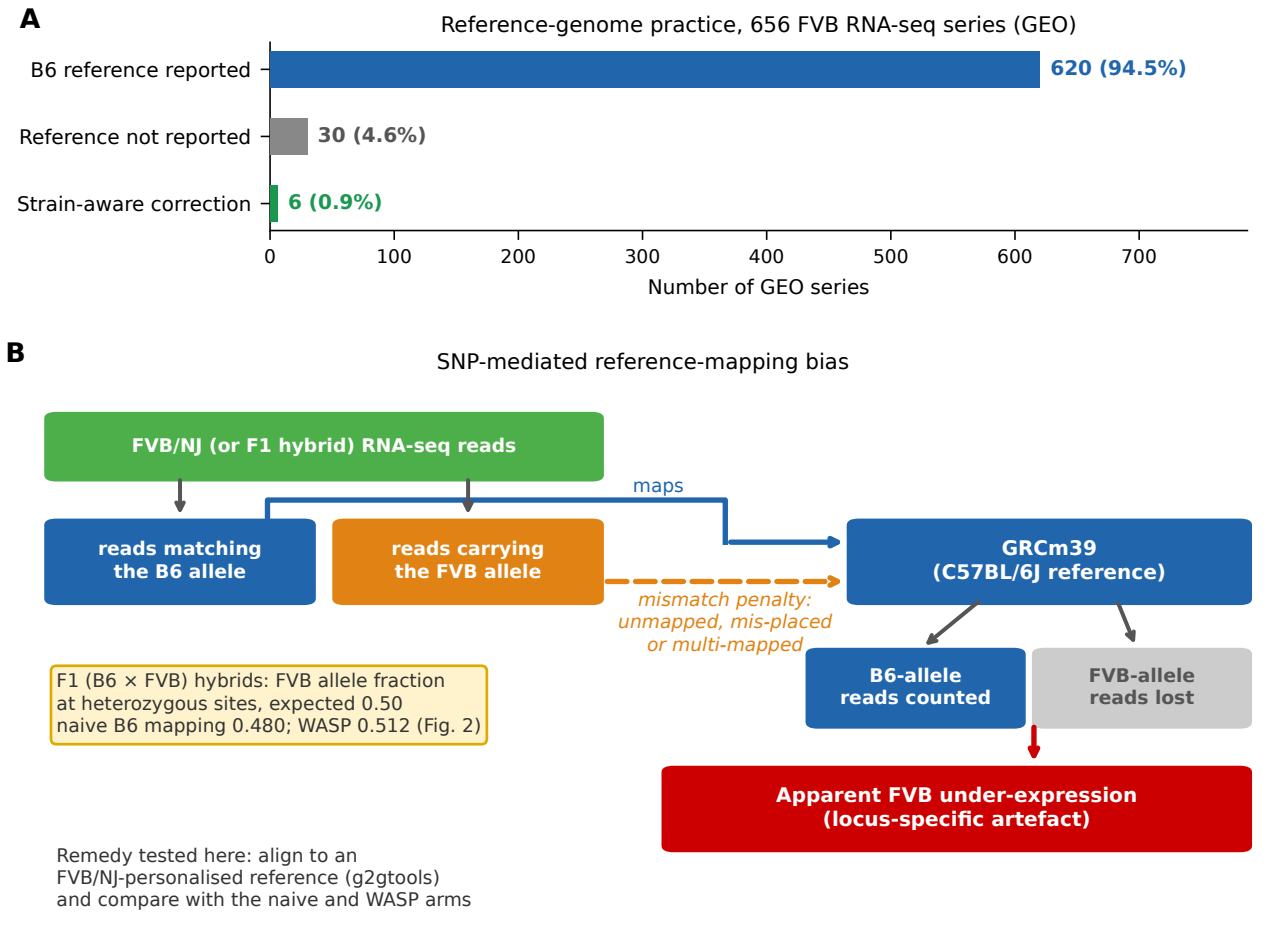

A
Reference-genome practice, 656 FVB RNA-seq series (GEO)
B6 reference reported
620 (94.5%)
Reference not reported
30 (4.6%)
Strain-aware correction
6 (0.9%)
0
100
200
300
400
500
600
700
Number of GEO series
B
SNP-mediated reference-mapping bias
FVB/NJ (or F1 hybrid) RNA-seq reads
maps
reads matching the B6 allele
reads carrying the FVB allele
GRCm39 (C57BL/6J reference)
mismatch penalty: unmapped, mis-placed or multi-mapped
B6-allele reads counted
FVB-allele reads lost
F1 (B6 × FVB) hybrids: FVB allele fraction at heterozygous sites, expected 0.50
naive B6 mapping 0.480; WASP 0.512 (Fig. 2)
Apparent FVB under-expression (locus-specific artefact)
Remedy tested here: align to an FVB/NJ-personalised reference (g2gtools) and compare with the naive and WASP arms